

In a SQL **SELECT** statement, clauses are **written in one order** but **executed internally in another order**.

SQL Query

```
SELECT column_name  
FROM table_name  
WHERE condition  
GROUP BY column_name  
HAVING condition  
ORDER BY column_name;
```

Actual Execution Order

1. **FROM**
 2. **WHERE**
 3. **GROUP BY**
 4. **HAVING**
 5. **SELECT**
 6. **ORDER BY**
 7. **LIMIT**
-

Explanation

1. FROM

First, SQL identifies the table.

FROM employees

2. WHERE

Filters rows before grouping.

WHERE salary > 50000

3. GROUP BY

Groups rows.

GROUP BY department

4. HAVING

Filters grouped data.

HAVING COUNT(*) > 5

5. SELECT

Chooses columns to display.

SELECT department, COUNT(*)

6. ORDER BY

Sorts final result.

ORDER BY department

7. LIMIT

Restricts number of rows returned.

LIMIT 10

Example

```
SELECT department, COUNT(*) AS total
FROM employees
WHERE salary > 50000
GROUP BY department
HAVING COUNT(*) > 2
ORDER BY total DESC
LIMIT 5;
```

Internal Execution Flow

1. FROM employees
2. WHERE salary > 50000
3. GROUP BY department
4. HAVING COUNT(*) > 2
5. SELECT department, COUNT(*)
6. ORDER BY total DESC
7. LIMIT 5